AIA · AGENTIC AI ANALYTICS · METHODOLOGY

The business-decision study
Study type 2: a cost model of options

How AIA takes a client's decision from *Zadání* to a report in which every number carries its source and its grade. Version 3, written bottom up from the ChemVitro exemplar.

THE SPINE · NINE STEPS

Zadání · *Data* · *Rešerše* · *Předpoklady* · *Model* · *Výsledky* · *Sociomapa* · *Osnova zprávy* · *Zpráva*

The client's rule is written at the first step and applied, as written, at the sixth. Every number in the report carries its source and one of seven grades. A method the register does not hold stays provisional.

Version 3 · 8 November 2026 · working version, for the product owner and the methodology owner
Replaces version 2 (36 pages, 5 November 2026), which recorded the methodology review R1–R10
Source: `prototypes/obchodni-model/methodology/` on branch `chore/mvp-r2-f` · built by `build.js`
Reference report: `prototypes/obchodni-model/exemplar/ChemVitro_2_model-lokality_nigel.pdf`, 44 pages, 5 November 2026
Case pack: `prototypes/case-chemvitro/` (branch `chore/case-chemvitro`) · Design canvas: `prototypes/obchodni-model/`

This is design, not product code. Section 10 says what exists in AIA today and what does not. Every client, site, person and number in the exemplar is fictional.

CONTENTS

What is in this document

What this is 3

1 The spine 4

1.1 The shape of each step chapter 4

1.2 The rules the whole spine keeps 4

2 Zadání: the decision and its rule 6

3 Data: the client's files under contract 8

4 Rešerše: public inputs with a state 10

5 Předpoklady: one table of inputs 12

6 Model: one computation for every option 14

7 Výsledky: the rule applied, and what could change it 16

8 Sociomapa: the futures map 18

9 Osnova a Zpráva: the outline, then the report 20

10 Built vs designed 22

10.1 The state on 8 November 2026 22

10.2 The plans that carry the build 22

10.3 Three honest statements 22

11 Perspectives, components and the two maps 23

11.1 Perspectives and components 23

11.2 The futures map 23

11.3 The weighting-scenario map (Mapa scénářů vah, board MV9) 23

12 Grades, sources and the release gate 25

12.1 The exemplar's seven grades 25

12.2 The four attributes behind a grade 25

12.3 Sources, numbered per chapter 25

12.4 The provisional state 26

12.5 The release gate (R9, proposed) 26

13 Governance 27

13.1 The validity review (R1) 27

13.2 Register entries the exemplar waits on 27

13.3 The methodologist's open questions 27

13.4 The owner's open decisions 28

13.5 Change propagation (K4) 28

A Glossary 29

B The exemplar in numbers 30

B.1 T1 · The rule at 8 % 30

B.2 T4 · Finance, base case 30

B.3 T6 · Operating cost per tonne in 2032, base case, EUR/t 30

B.4 T7 · Relations between sites across the futures 30

B.5 T8 · Shared failure of the two leading sites 31

B.6 The decision boundary and the input table in brief 31

C Board index 32

BEFORE THE CHAPTERS

What this is

On 8 November 2026 the executive chose report 2 of the ChemVitro case study as the report AIA must produce. This document is the methodology behind that report, step by step, and the measure the MVP will be held to.

The three outcomes of the pivot

Outcome	Where it lives	Who
One MVP prototype	The Design canvas *AIA Obchodní model*, `prototypes/obchodni-model/`; thirty boards, Czech, one path from *Zadání* to *Zpráva*	sessions A–E
One methodology PDF	This document, `prototypes/obchodni-model/methodology/`	session F
One MVP scope	`prototypes/obchodni-model/MVP-SCOPE.md` and board MV0; the methodology and the scope must agree	session A

What the MVP produces

Study type 2, *Model nákladů lokality*: a decision between options, judged by a rule the client wrote before any result, on a cost model computed by code, with the uncertainty of every input carried through to the verdict. The exemplar is a type-2 report on a fictional client, ChemVitro, choosing among eight sites for a chemical plant or keeping its supplier. Its parts carry the ids R2-P01 to R2-P18 and its tables T1 to T9; this document cites them by those ids.

Types 1 and 3 of the case pack (the situation report without a model; the model beside a worker survey) are not the MVP. The survey lane is frozen (ADR 0046, decision 2).

How to read it

- Chapter 1 is the spine: the nine steps in one table.
- Chapters 2 to 9 take one step each, all in the same shape: what goes in, what comes out, the checks, how the exemplar did it, what AIA has today, what is new, and what the ChemVitro acceptance expects.
- Chapter 10 says, truthfully, what is built and what is designed. No type-2 code is on `develop`.
- Chapters 11 to 13 cross the steps: the business perspectives and the two Sociomaps; grades, sources and the release gate; governance.
- The appendices hold the glossary, the exemplar in numbers and the index of the canvas boards.

Czech words in italics are terms a person sees on AIA's screens or in the report. English words in `code` are files, modules or plan chunks in the repository. "The review" is the product owner's methodology review of 5 November 2026 (ADR 0046 § The methodology review, R1–R10, K1–K6). "The answer key" is `prototypes/case-chemvitro/ANSWER-KEY.md`. "The owner" is the product owner; "the methodologist" is the methodology owner, A. Bahbouh.

> **Three things this document does not do.** It does not invent a number: every ChemVitro figure comes from the exemplar's tables or the case pack. It does not decide what waits on the owner or the methodologist; chapter 13 lists those items. It does not describe how a screen works: the boards do that.

CHAPTER 1

The spine

A type-2 study is nine steps. The client's rule is written at the first and applied, as written, at the sixth. Everything between them exists to make that application honest.

Step	Goes in	Comes out	The check	In the exemplar
1 ***Zadání***	the decision, its options and constraints, who decides and when, the client's files	the rule with its conditions, frozen and dated; interpretations recorded	frozen before any comparative result; every condition has a threshold and a measure	eight sites and the buy baseline; the CFO's rule of 2. 11. 2026 with four conditions; one interpretation waits for the CFO (R2-P02)
2 ***Data***	every client file and document	a contract per file; a findings table with a disposition per finding	units, currencies, price years, duplicates, missing, outliers, quotes that do not cover the same thing	42 findings; the key one is the Hanshin price (R2-P04 § 3.4, P1)
3 ***Rešerše***	what the files cannot give: public prices, rates, taxes, limits, law	typed public inputs, each with a quote, a source, a date and a state	verified at the source, or marked *prozatímní*	16 of 54 inputs public and unverified; *Stav podkladů* says so on page 1 (R2-P01)
4 ***Předpoklady***	the contracts, the public inputs, the client's answers	one table of inputs: low, likely, high, distribution, source, grade, guarantor, state	estimated from client data where possible; the scope of every estimate checked; correlations declared with a stability test fixed in advance	54 inputs (T9); overrun and delay from 36 comparable plants, drawn together (R2-P05)
5 ***Model***	the input table, the options, the timeline	the base case and the seeded futures; the same bytes on a rerun	one model for every option; a declared stopping rule; one change at a time in the extra computations	cost of supply 2027–2044; 20 000 futures, seed 20270201 (R2-P04 § 3.3, Appendix B)
6 ***Výsledky***	the futures and the base case	the rule applied as written; what could change the verdict; value of information; finance, operations, schedule	the verdict shown as a boundary; no probability outside the model; information valued for the rule first	no site passes; the flip at 14/15/16 % compensation chance; EVPPI second (R2-P06–P11)
7 ***Sociomapa***	the options' results across the futures, or their weighted scores across scenarios	a map with a mode contract, its relation matrix and what it may claim	links only where the contract allows; a stability test declared in advance; relative apart from absolute	the futures map failed its stability test and the report says so (R2-P12, T7, T8)
8 ***Osnova zprávy***	every claim the report will make	an outline ordered by the argument; each point with its source or method and why it matters	no point without a source; a point that is a judgement is marked as one	the part ids R2-P01…P18 and the grade legend (R2-P04 § 3.5)
9 ***Zpráva***	the closed outline	the report: summary, chapters, limits, literature, appendices, tables	every number with a grade in small type and a numbered source at the chapter's end; a validity review before approval	44 pages, seven grades, *Meze studie* admits what is unverified (R2-P15)

1.1 The shape of each step chapter

Chapters 2 to 9 use seven parts, always in this order and under these names, so that two steps can be compared line by line.

Part	What it answers
What goes in	The inputs the step takes, and from which earlier step or from the client.
What comes out	The artefact the step produces, as the next step and the report read it.
Checks	What must hold before the step is closed. A check is a refusal or a record, never advice.
How the exemplar did it	The ChemVitro report's own practice, cited by part id and table.
In AIA today	The code on `develop` the step reuses, by module. Nothing listed here is type-2 code; it is what the pivot stands on.
New	What has to be built, with the board and the plan chunk that design it where one exists.
ChemVitro acceptance	What the answer key planted for this step, and which of the review's criteria K1–K6 the step answers.

1.2 The rules the whole spine keeps

- **The client's rule decides.** It is written before any result and judged as written (K3). Anything else the study computes is evidence for "what could change the conclusion", not for the recommendation (review R3, R10).

- **A number is a reference to its origin.** It carries a source, a method, an uncertainty and a validation status (review I3). A number the origin does not support is not printed.
- **Unknown is never good.** A missing input keeps a range; a gate that is unknown blocks the recommendation until it is evidenced (review R2). A probability is a share of the model's futures, never a claim about the world.
- **Models propose, code computes, a person approves.** AIA's standing rule (ADR 0046 decision 1). A language model drafts text and suggests ranges; every number in the report comes from a program that gives the same bytes on a rerun.
- **A method the register does not hold stays provisional.** Its results print *odhad (metoda čeká na schválení metodikem)*, and a recommendation that rests on it stays *předběžné doporučení* (review R9, proposed; chapter 12).
- **Changes propagate.** A revised input marks every downstream result and approval stale until someone reviews it (K4). Every extra computation changes one assumption at a time and says so (R2-P15).

CHAPTER 2 · STEP 1

Zadání: the decision and its rule

The study begins by writing down what will be decided, between which options, under which constraints, by whom and when, and by which rule. The rule is frozen before anyone sees a comparison of the options.

2.1 What goes in

- The client's brief, as typed or pasted: one field of free text and the files that come with it (the case pack's brief is in `CASE.md` § 3).
- The decision-maker and the date the decision is taken, and the date the study must be ready.
- The options, each with the client's source for it, and a baseline every option is compared with. For ChemVitro: eight sites from the agencies' offers and "keep buying" from the supplier's letter.
- The constraints every option must meet (investment ceiling, start date, one plant at one site).
- The client's answers to the study's questions. The case pack fixes them in `CASE.md` § 8; where an answer is "we don't know", the input becomes a judgement with a range and says so.

2.2 What comes out

- **The decision record:** what is decided, why now, the goal, who decides, by when. The exemplar's § 1.1–1.2 (R2-P02).
- **The options and the baseline**, named as the client names them, with the source of each.
- **The rule**, frozen and dated, in the client's words: the measure it reads, the choice it makes, and each condition with its threshold. ChemVitro's: the highest median saving against buying, NPV of the cost of supply 2027–2044 at 8 %, if P(beats buying) ≥ 0,75, P(loss > 20 mil. EUR) ≤ 0,10, P(production by 1. 7. 2030) ≥ 0,80 and investment ≤ 175 mil. EUR; else recommend nothing and name the assumption most worth checking (R2-P02 § 1.3).
- **Interpretation records.** Where a condition does not say how it is measured, the study writes its reading down, the alternative reading, what each would change, and who confirms it. The exemplar's condition (4) names no probability; it is judged on the base case, the stricter reading "in every future" would fail all eight sites without changing the verdict, and the CFO is asked to confirm (R2-P02 § 1.3).
- **The views the client asked for beside the rule:** after-tax NPV, the verdict at 6 % and 10 %. They are shown; the rule does not read them.
- **The analyses the report will carry** (board Z4) and the perspectives that apply to the case (chapter 11).

2.3 Checks

- The rule exists before the first comparative result, and the freeze is dated and attributed. Reading the client's files at intake is feasibility work and is recorded as such (review I2).
- Every condition has a measure the model can compute and a threshold. A condition without one gets an interpretation record, never a silent default.
- An amendment of the rule after the freeze is approved by the decision-maker, dated, and printed in the report (board Z3, *zámek*).
- Every option is described comparably: the same plant, the same horizon, the same discount rate (review R1, question 1).
- The brief is Class B material by rule (`domain/material_rules.py`); a language model may read it, a client's dataset is Class A and is not sent to a model.

2.4 How the exemplar did it

The report states the decision, the eight sites, the fixed plant (40 000 t a year, 95 FTE, one process design) and the dates: the board meets on 29. 1. 2027, the site is decided on 1. 2. 2027, production by 1. 7. 2030, investment at most 175 mil. EUR (R2-P02 § 1.2). It prints the rule in full and says why its date matters: "that the rule was written before the computation prevents the result from being justified afterwards by a rule that would suit it" (§ 1.3). It records the one interpretation, prices the alternative reading and asks the CFO. It also notes that the rule reads the pre-tax NPV and that after-tax figures and other rates are shown at the client's request.

2.5 In AIA today

- The study's working copy with revision-checked saves and the operation log (`application/working_copy.py`, `domain/edit_ops.py`, `infrastructure/study_workspace_repository.py`), and immutable Design Revisions (`domain/design.py`, `infrastructure/study_design_repository.py`). The frozen rule is a revision of this kind.

- The Brief stage of the research workbench (`apps/web/src/components/rehome/research/`, `BriefStep`) and its materials (`domain/brief_materials.py`), reused as the screen the brief is typed into.
- The Lead, which triages a brief and proposes edits a person accepts (`application/research_lead.py`, ADR 0026, ADR 0041).
- The approval ledger with its closed set of subjects (`domain/approval_ledger.py`): the rule's freeze and every amendment become ledger rows.
- Material classes by rule (`domain/material_rules.py`, ADR 0008).

2.6 New

- A study kind for a business decision, `StudyKind.BUSINESS_MODEL` (ADR 0044; `cost-model-study` CP3), and the six sub-steps of *Zadání* on boards Z1–Z7: start, triage, the decision core with the rule, the analyses, the model's parts, readiness.
- The rule as a typed object with conditions, thresholds and the measure each reads, frozen as a Design Revision and judged as written (`cost-model-study` CR7, board Z3 in revision, session B).
- Interpretation records as a type: reading, alternative, effect, who confirms (new; board Z3).
- The perspectives chosen for the case (chapter 11), each with its components, in *Zadání* rather than in the model.

2.7 ChemVitro acceptance

- The rule is the CFO's of `CASE.md` § 6, with all four conditions and the "recommend nothing" clause, dated before any result (MV0 M-4; K3).
- The investment condition has an interpretation record, and the record names the CFO as the person who confirms it.
- The baseline is "keep buying" at the supplier's terms, not a site.
- The study asks the client only what `CASE.md` § 8 answers; anything else is a judgement with a range, labelled *úsudek*.

CHAPTER 3 · STEP 2

Data: the client's files under contract

Every client file is read, checked and bound by a contract before a number from it reaches the model. The checks produce findings, and every finding gets a disposition the report can print.

3.1 What goes in

- The client's data files: for ChemVitro ten files in `data/`, from monthly purchases to air-quality measurements (`CASE.md` § 9). The worker survey belongs to type 3 and is not read.
- The client's documents: seven PDFs, from the board's brief to the consultant's memo on air permitting. Each number taken from one is cited with its page.
- The decision record from step 1, which says what each file is for.

3.2 What comes out

- **A data contract per file:** which columns are used and how; units, currency and price year per column; which columns are personal and never leave the analysis copy; the period covered (board MV1; `client-data-in-analysis` chunk 2).
- **A quality report per file**, accepted by a person before use: missing, duplicate and out-of-range counts per used column over every row (`client-data-in-analysis` 2a).
- **The findings table with dispositions.** Every finding of the checks, with one of four dispositions: used; corrected, with the reason; left out; taken over as an assumption with a range. The findings that change the result or its reading go into the report (the exemplar's P1, R2-P18 Appendix B).
- **A document register:** each document by the name the report cites it under and its file name (the exemplar's Appendix A).

3.3 Checks

The checks are the exemplar's list (R2-P04 § 3.4), made explicit:

Check	What it looks for	ChemVitro instance
Units	a row whose unit differs from its column's	July 2025 volume 2 844 000: kilograms, read as 2 844 t (P1)
Currencies	mixed currencies in one column; a currency column that contradicts the header	comparable plants in EUR, CZK and PLN; one purchase row in USD (answer key § 3)
Price years	amounts in different years' prices without a year	one comparable plant without a price year (P1)
Duplicates	the same entity twice	one comparable plant entered twice; counted once, 36 plants (P1)
Missing values	cells the contract expects and does not find	no start date of the energy price fixation; waste prices only to 2028 (R2-P01 *Stav podkladů*)
Outliers	a value outside the range the file itself declares	one plant with capacity 91 kt against 18–60 kt; used only for overrun and delay (R2-P15)
Scope of a quote	quotes that do not cover the same thing	a cheaper PL2 waste quote covers only non-hazardous waste; PL1's rail rate assumes a spur that does not exist (answer key § 3)
File against brief	a number the brief states and the file contradicts	"about 1 470 EUR/t" against a 2026 average of 1 694 and 1 868 in September 2026 (P1)
Personal data	a column holding a person's identifier	none in type 2's files; the survey's respondents are pseudonymous and belong to type 3

A contract that names a column the file does not hold, or a quality report the person has not accepted, blocks the file from the model (MV0 M-1).

3.4 How the exemplar did it

The model uses every client file except the worker survey and all seven documents (R2-P04 § 3.4). The checks produced 42 findings, each with a recorded decision; the ones that change the result are in table P1 (Appendix B): the Hanshin price modelled as a range 1 470–1 694–1 868 EUR/t; the kilogram row; the single spot purchase of February 2026 turned into a spot premium of 20–60 %, most likely 39 %; the duplicate, the missing price year and the 91 kt outlier among the comparable plants; consumption growth of 4 % in the data against the client's 1–3 %, the client's range used; raw-material freight not added because the raw-

material price is delivered to site; Odrzyca's rail quote and Eschfeld's bridge. Twenty-two inputs the files do not hold became judgements with a range. A separate document, *Kontrola dat*, holds the contracts and all 42 findings (Appendix C); the report prints the ones that matter.

3.5 In AIA today

- A client's sources kept once per client and SHA-256, read by a format reader into segments with locators (`domain/client_sources.py`, `infrastructure/source_reading.py`, `infrastructure/client_source_repository.py`): PDF by page and table row, XLSX and CSV by sheet and row, DOCX by paragraph.
- The dataset preview: per column the header, kind, fill, distinct values and samples, with personal columns detected and masked (`domain/client_datasets.py`; patterns pinned in `fixtures/client_datasets/personal.json`).
- The intake sheet, *Přidat podklady*: each file kept and read at once, one knowledge item per file (`apps/web/src/components/aia/clients/IntakeSheet`).
- Document text with ZIP bounds and a bounded table reader (`infrastructure/document_text.py`).
- The evidence policy for client data: client-reported evidence under an accepted contract (ADR 0027, ADR 0037; `client-data-in-analysis` 8a built).

3.6 New

- The data contract as a type and the route a person accepts it through, the quality report, and the analysis copy that holds only the contract's columns (`client-data-in-analysis` 2, 2a, 3, 3a, 3b; board MV1 in revision, session B).
- The findings table with its four dispositions, kept with the contract and printed in the report (new; board M1; `cost-model-study` CR1 covers reconciliation, period coverage, currency and robust outliers, not the disposition record).
- Per-column unit, currency and price year in the contract, so a conversion is a declared step and not a reading (board MV1).
- The document register with cited names (new; the exemplar's Appendix A).

3.7 ChemVitro acceptance

The answer key planted these in the files (§ 3). A study that misses one has a gap in its checks, whatever its verdict.

- `srovnatelne_zavody.xlsx`: P-17 twice, the second with a rounded investment; P-23's capacity 91 in t/day, not kt a year; P-31 without a price year; investments in EUR, CZK and PLN in each plant's own price year.
- `nakup_vtx40_2023-2026.xlsx`: the 2025-07 volume in kg; the 2024-03 price in USD; an extra 2026-02 row from another supplier, the spot price for a late start. The key names 2 180 EUR/t; the exemplar read the spot purchase as 2 320 against 1 672 EUR/t (P1). The acceptance reads the file, not either text.
- `trh_prace_regiony.xlsx`: wages gross, in local currency, without employer contributions.
- `odpady_nabidky.csv`: PL2's second quote covers only non-hazardous waste.
- `doprava_nabidky.csv`: PL1's rail rate assumes a spur to the site.
- `hodnoceni_lokalit.csv`: one rater scored with the scale reversed and says so in a note; the consultant adds about 2,5 points to the two sites its agency brokered; two cells are missing. Type 2 does not use this file in its model; if it draws an expert map, the map must show these.

Criteria: K6, claim integrity, where a table and the text disagree about a reading; K4, since a corrected finding propagates to every result that read it.

CHAPTER 4 · STEP 3

Rešerše: public inputs with a state

What the client's files cannot give, public sources give: prices, rates, taxes, limits and law. Each is a typed input with a quote, a source and a date, and it stays provisional until someone has verified it at the source.

4.1 What goes in

- The list of public inputs the model needs, from the decision record and the contracts. For ChemVitro, `CASE.md` § 10: the EU ETS allowance price and its outlook and whether a 24 MW boiler falls under the scheme; air-quality limit values and the permitting rules of three countries; hazardous-waste fees; wages, employer contributions and unemployment; corporate income tax, depreciation and the nature of each incentive; industrial energy price outlooks; exchange rates.
- The client's data posture (*Pravidla*, `domain/client_rules.py`) and the study's budget, which bound what a priced route may spend.
- Findings that frame the problem: the literature the report cites (R2-P03), found and quoted the same way.

4.2 What comes out

- **Typed public inputs** (board MV2): a value with its unit, the quote it was read from, the source and its date, the kind of source (*Úřední / Oborový / Ostatní*), the grade (*veřejný zdroj* or *právní předpis*) and the state: *ověřeno* once checked at the source, else *prozatímní*.
- **Findings by analysis** (board MV2b): PESTEL, Porter or benchmark findings as proposals a person accepts, each a quote in a captured source.
- **The literature review** as the report prints it: for each work, what it says and what follows for this report (R2-P03, five works).
- **The state line for the report's first page:** how many inputs are public, how many unverified, and whether the unverified ones could change the verdict (R2-P01 *Stav podkladů*).

4.3 Checks

- A finding is a quote in a captured source, or it is nothing (ADR 0017). A value without a quote and a date is not an input.
- A query's data class is inherited from what it was written from and never lowered by its words (ADR 0017, `domain/deep_research/classification.py`). A client's dataset never leaves as a query.
- A provision is cited; no legal advice is given (`CASE.md` § 10, R2-P04 § 3.3).
- An unverified input is marked *prozatímní* in the input table, in the chapter that uses it and on page 1. Verification at the source is a dated act by a named person and changes the state, nothing else.
- A public input the exemplar's method tests for sensitivity says whether its range changes the verdict (chapter 7).

4.4 How the exemplar did it

Sixteen of the 54 inputs are public: the allowance price (89,10 EUR/t on 28. 10. 2026, Investing.com, unverified), its growth from December 2030 futures, free allocation, energy price volatility and long-term movement, three price correlations, the CZK and PLN rates, employer contributions in three countries, inflation (T9). The report says, on page 1, that the authors could not verify them at the source because access to public sources was closed while the report was written, that the amounts may change after verification, and that the correlations and the energy trend do not change the verdict while stricter 2030 air-quality limits could (R2-P01). The tax rates, depreciation, the form of incentives and Directive (EU) 2024/2881 are likewise provisional; the incentives are read correctly as tax reliefs in Czechia and Poland and a cash grant in Germany (R2-P10). The literature chapter cites AACE 18R-97, Flyvbjerg, Merrow, Vose, Strong and Oakley, Williamson and Lempert, Popper and Bankes, each with the consequence for the report (R2-P03).

4.5 In AIA today

- Deep Research, frozen since #166: plan, investigate, merge, verify, synthesize, publish; grounding of every quote in a content-addressed snapshot of the same track (`domain/deep_research/`, `grounding.py`, `web.py`, `sources.py`); the sealed bundle.
- The one way a query or URL leaves: `application/web_retrieval.py` `RetrievalGate` (classify, egress, metering, reserve, journal, call, journal). Live routes: Czech and English Wikipedia, priced Brave, Common Crawl (`aia_executors.deep_research_runtime`).
- Design Research and Interpretation Research as the two purposes; a finding never writes a design or a deterministic artefact (ADR 0021).

- A captured source a person checked becomes a SOURCE knowledge item, *Ověřeno, přidat zdroj* (`application/verified_sources.py`).
- The Lead's web lookup, accepted and not built (ADR 0040).

4.6 New

- Typed public inputs: a value read from a quote by code, with unit, date, source kind and state, as a provenance of a model input (`cost-model-study` CP1 names `public_source`; board MV2 in revision, session B).
- The state *prozatímní* / *ověřeno* on every public input, the dated act of verification, and the first-page state line composed from the input table (new).
- Public routes beyond Wikipedia and Brave for registers and statistics (named on MV0 as `public-data-routes`; no plan file in the repository yet).
- Findings by analysis (PESTEL, Porter, benchmark) as proposals, each a quote (board MV2b, unchanged in this round).

4.7 ChemVitro acceptance

- Every public value in the model carries a quote, a source, a date and a state; none is invented (`CASE.md` § 11).
- The 24 MW boiler is above the 20 MW threshold: carbon is a cost at every site (answer key § 3, memo p. 1).
- Incentives are read by their nature: the Czech investment incentive and the Polish CIT exemption are tax reliefs and do not change the pre-tax figure the rule reads; only the German GRW is a cash grant. Subtracting PL2's "CIT exemption up to 50 %" from the investment is a misreading (answer key § 3).
- The revised directive (EU) 2024/2881 with its 2030 limits is noted as a caveat, as the exemplar does; it is not planted and earns credit, not a requirement.
- Criteria: K2, where a public range is a decisive input; K5, where verification at the source has a date that the plan of action depends on.

CHAPTER 5 · STEP 4

Předpoklady: one table of inputs

Every input of the model is one row of one table: its low, likely and high value, its distribution, its source, its grade, the person who stands behind it and its state. The table is the contract between the data and the model, and the report prints it whole.

5.1 What goes in

- The data contracts and the findings table from step 2; the typed public inputs from step 3; the client's answers from step 1.
- The comparable cases the client holds, from which ranges can be estimated rather than judged: for ChemVitro, 36 comparable plants with their overrun, delay and what each built beyond the plant.
- The scope statement of every estimate the client supplies: what the EPC contractor's figure includes and excludes (`epc_odhad.pdf` p. 3).

5.2 What comes out

The input table, the exemplar's T9 (R2-P18). Its columns are the method:

Column	Rule
Vstup, jednotka	one input, one unit; a fixed parameter is a row like any other
Dolní · nejspíš · horní	three values; equal for a fixed value. What the ends mean is written down for every decisive input (review R6): a 90 % interval of an estimate, contractual bounds, the worst and best credible case
Rozdělení	PERT by default for a range a person stands behind; the estimate's own distribution where it comes from data; an empirical draw from comparable cases; yes/no with a probability for an event; a mixture where experts disagree; a fixed value (*pevná hodnota*)
Zdroj	the file and row, the document and page, the public source and date, or "not in the files"
Stupeň	one of the grades of chapter 12: *z dat klienta*, *odhad*, *úsudek*, *veřejný zdroj*, *právní předpis*
Stojí za ním	the guarantor: a named role in the client (the CFO, head of purchasing, head of EHS) or the author. Every assumption has an owner (the review's "to keep")
Stav	*ověřeno* (from client data), *prozatímní* (public, unverified) or *úsudek* (the client has no answer)

Beside the table: **the correlation record** (which inputs move together, with what coefficient, why, and the stability test their judgement will be put to), **the scope check** of every supplied estimate against each option's documents with the exclusions priced, and **the stress cases** outside the ranges, kept apart from the futures (board M2 § 04).

5.3 Checks

- Where the client's data allows an estimate, the range comes from the data, not from a judgement; the row says which. The exemplar: overrun and delay from the comparable plants, not from the estimator's ±25 % (R2-P05).
- Two things that happened together in the data are drawn together in the model. Overrun and delay have a Spearman correlation of 0,89 across the comparable plants, so each future draws one real plant and takes both (R2-P05).
- Every supplied estimate is checked for scope against each option's documents, and each exclusion is priced or stated as missing (R2-P05, fourth assumption; answer key § 3: a study that adds nothing to the EPC figure has missed all three traps).
- A decisive input has an uncertainty type: natural variation, parameter, expert disagreement, model structure, or a severe event outside the range; the last goes to a stress test, not into the distribution (review R6; board M2 § 02).
- Experts who disagree from the same documents are a mixture that does not narrow the range; the report says they differ in reading, not in facts (board M2).
- A correlation that is a judgement is declared as one, with the alternative settings it will be tested under, before anything is computed (R2-P05 fifth assumption, Appendix B).
- No input the files do not hold is given a point value. It gets a range and the grade *úsudek*; the exemplar has 22 such inputs (R2-P18 Appendix B).

5.4 How the exemplar did it

Chapter 4 (R2-P05) names the two assumptions the verdict rests on that no document supports: the 20 % chance that an authority imposes compensation measures where the station is under the limits, and the terms of purchase after 2035. Then three more: overrun and delay from 36 comparable plants (median overrun 17 %, 90th percentile 30,2 %, maximum 61 %; delay median 3

months, maximum 15), drawn jointly; the exclusions of the EPC estimate priced per site from the site's documents (18 ha of land everywhere; the bridge at Eschfeld 3,2–5,0 mil. EUR; the spur at Odrzyca 3,4 km at 1,9–2,8 mil. EUR per km; pre-treatment at Lipová and Lindenau 6,7–14,1 mil. EUR; compensation 3–15 mil. EUR where imposed; the park fee at Wendelstedt 1,1 mil. EUR a year); and the energy price correlations as judgement (0,8 electricity between countries, 0,6 electricity–gas, 0,4 gas–allowance), with the stability test declared in Appendix B. The base case already carries the median overrun. T9 lists all 54 inputs with every column above.

5.5 In AIA today

- Nothing holds an input table yet. The nearest ancestors are the simulation core's frozen WorldModel with reject-not-clip validation (`domain/simulation/`) and the research template's readiness rules (`domain/research_design.py`), which show the pattern: a typed object, validated, pinned to a run.
- Guarantors and audit: the approval ledger (`domain/approval_ledger.py`) and the audit trail every admin route writes.
- Methodology as data the methodologist owns: versions, activation and bounds per editable field (`domain/methodology/`, ADR 0023). The distribution defaults and the stability thresholds belong there.

5.6 New

- The input as a platform type with provenance (`cost-model-study` CP1: `client_dataset`, `stated`, `public_source`) and ranges with a guarantor on every input (CR2); the input table as the stage *Předpoklady* (board M2 in revision, session B).
- Distributions by source and the correlation record: PERT, Iman–Conover rank correlations, a non-PSD matrix refused (CR3); plus what CR3 does not name: an empirical joint draw from comparable cases, a yes/no event with a probability, a mixture for expert disagreement (review R6; board M2 § 02–03).
- The estimate from comparables as a method of its own: the client's cases filtered by the contract, the statistic computed by code, printed with the sample size (new; the exemplar's third assumption).
- The scope check: a supplied estimate's inclusions and exclusions as data, each exclusion matched to each option and priced or marked missing (new).
- The state column and its three values, read from steps 2 and 3 (new).

5.7 ChemVitro acceptance

The answer key's three traps live in this step. Each is a cost the brochure hides and a site document reveals; the study must find and price it.

Trap	Where it is	What the exemplar priced
Wastewater, CZ2 Lipová and DE2 Lindenau	`vyjadreni_cov.pdf` p. 2 and 5: the sewer bylaw's sulfate and COD limits against the process effluent (1 800 and 2 400 mg/l); Lindenau's plant has spare capacity for 250 m³ a day against 430 needed	pre-treatment 6,7–14,1 mil. EUR at both, from 8 comparable plants; Lindenau's direct-discharge permit 0–9 months (T9)
Air, CZ1 Vrbice and PL2 Bory	`ovzdusi_stanice.csv` 2025 values over the limits, combined with `memo_povolovani_ovzdusi.pdf` p. 2–3: compensation measures and 9–15 / 12–18 months longer permitting	compensation with probability 0,8, 3–15 mil. EUR, 9–18 months (T9); both sites then miss the deadline in most futures (T1)
Logistics, PL1 Odrzyca and DE3 Eschfeld	`vyjadreni_doprava.pdf` p. 7: the investor builds the 3,4 km spur and buys the land; p. 6: the bridge carries 16 t, tankers weigh 40 t	the spur at 1,9–2,8 mil. EUR per km plus road haulage until built; the bridge 3,2–5,0 mil. EUR and 12–30 months to heavy-traffic approval (T9)

- Overrun from the comparables, median 17,0 %, not the estimator's band; the duplicate removed first (answer key § 3).
- The decoys left alone: Kamenec's single 2021 exceedance (under the target since); Eschfeld's plant charges from COD 2 500 mg/l against 2 400 in the effluent; Lipová's turning lane at about 2 mil. Kč; Bory's municipal plant does accept the effluent.
- Wendelstedt's park fee of about 1,1 mil. EUR a year, named only in the wastewater statement (p. 4).
- Criteria: K2, uncertainty fitness, is this step's criterion: decisive distributions, dependence and stress cases justified. K1's second question (material costs and dependencies in the model) is answered by the scope check.

CHAPTER 6 · STEP 5

Model: one computation for every option

The model computes the measure the rule reads for every option in the same futures. It is code, pinned by version and seed, and a rerun gives the same bytes. Nothing in it is a forecast or a legal opinion.

6.1 What goes in

- The input table of step 4, frozen as a revision; the options and the baseline of step 1; the rule's measure.
- The timeline every option shares: for ChemVitro, decision and permit applications on 1. 2. 2027, permitting 8–11 months plus the site's extensions, construction 22 months plus the drawn comparable plant's delay, ramp-up at 60–90 % for six months (Appendix B).
- The seed and the number of futures, or a stopping rule that sets the number.

6.2 What comes out

- **The base case:** one computation with every input at its likely value. It compares the options and feeds the sensitivity analyses; it is not an estimate of what the client will save (R2-P10).
- **The futures:** N seeded draws of all inputs at once, with their correlations, each future computing the baseline and every option. From them: the median, the 90 % interval, and every probability as a share of futures (R2-P04 § 3.3).
- **The extra computations** on the same model and the same futures, each changing one assumption: the decision boundary over an assumption's range, delay of the decision, hiring delay, alternative terms, correlation settings (Appendix B).
- **The run record:** model type and version, input revision, seed, N, the methods pinned, the artefacts with their SHA-256 (MV0 M-2, M-3).

6.3 Checks

- Every option is computed in the same futures by the same model; the baseline is one of them (R2-P04 § 3.3).
- The same inputs and seed give the same bytes (M-3). This verifies the implementation, not the model's fitness for the decision (review R1; chapter 13).
- The number of futures follows a stopping rule tied to the decision: the verdict of each condition is known to within a stated margin (board M2: ±0,5 pp on the pass probabilities) or N is declared and justified. The exemplar declares N = 20 000 (R2-P04 § 3.3).
- Cash flows are discounted at the rule's rate, nominal or real as declared, with the convention named (mid-year, inflation 2 % provisional; Appendix B).
- Tax is inside the model as a fixed parameter per country, so the after-tax view agrees with every other view (review I4). The rule reads the pre-tax figure.
- What is outside the model is listed where the model is described, not discovered in the limits (R2-P15: transfer prices, hiring time in the schedule, raw-material freight, the annual step).
- An extra computation changes one assumption; combinations that are not computed are said not to be (R2-P15).

6.4 How the exemplar did it

The cost of supply is the investment, operations from start-up, VTX-40 bought until start-up, purchases above 40 000 t, less a residual value of 30 % of the investment in 2044; the saving of a site is the baseline's cost less the site's, in NPV, positive when the plant is cheaper (R2-P04 § 3.3). Purchases follow the supplier's formula to 30. 6. 2030, then the renewal premium, or the spot price while the plant is late (Appendix B). Operations are six items per tonne: raw materials at 540 EUR/t delivered, electricity and gas at the sites' offers moving with the country's price path, CO_2 at the allowance price less free allocation, water, sewer, hazardous waste and product freight at the quotes, 95 staff at the operator wage times a wage mix and contributions, maintenance and insurance at 2–4 % of the EPC estimate (Appendix B). The futures number 20 000 with seed 20270201. The base case carries the median overrun of 17 %; the futures draw one comparable plant for overrun and delay together. Eight extra computations reuse the same futures, each changing one assumption (Appendix B).

6.5 In AIA today

- Runs as workflows on the worker, with lease-fenced steps, artefacts, provenance and dependency edges, each artefact pinned by SHA-256 (`apps/worker`, `infrastructure/artifact_repository.py`, `domain/workflow.py`, `domain/pipeline.py` fingerprints).

- A run pins the organization's methodology in force and records it in its identity (`domain/methodology/pins.py`, `application/research.py`).
- The deterministic simulation core: a frozen WorldModel, reject-not-clip validation, seeded scenarios and frozen results (`domain/simulation/`). Its discipline is the model's; its content is not.
- Seeded synthetic data from `random.Random(seed)` and the Sociomap golden fixtures pinned by SHA-256: the rerun-equals-bytes practice (`packages/aia_core/tests/fixtures/sociomap/`).
- Budgets and cost ceilings per run, which a model computed by code does not need (a CALCULATED run has a known 0,00 USD ceiling, `cost-model-study` CP5).

6.6 New

- The platform: quantity types, the model-type contract with a registry and a run's pin of type and version, the business-model workflow with inputs frozen by artefact id, the API and the stage screens (`cost-model-study` CP1–CP7; boards M3, M4 in revision, session C).
- **A model type for a type-2 study:** a cost of supply (or cost of operation) over a shared timeline with permitting, construction, ramp-up, a baseline bought until start and a residual value. The planned type *scénáře nákladů* (CM1–CM7) has cost lines, drivers and up to twelve scenarios without a timeline or a baseline; type 2 needs the timeline. Whether this is a second model type or an extension of the first is a design decision for sessions A and C.
- Monte Carlo with the input table's distributions and correlations, the joint draw from comparables, the stopping rule, and PCG64 under the run's seed (CR3; board M3).
- The extra computations as declared operations on a stored run: a boundary over one input's range, a shift of the timeline, an alternative terms table, the correlation settings (new; board M4).
- NPV, IRR and payback at a stated rate (CM7, `quantitative-families` FIN2), and a tax parameter per country inside the model.

6.7 ChemVitro acceptance

- Seed 20270201, 20 000 futures, the same seed gives the same result (`STUDIES.md` § 2, MV0 M-3).
- The spot price for a late start: any VTX-40 needed after 30. 6. 2030 and before the plant runs is bought at spot, as in February 2026 (`CASE.md` § 8).
- Tax reliefs out of the pre-tax figure the rule reads; after-tax at each country's rate (answer key § 3).
- Every option on the same timeline from 1. 2. 2027; a site's extra months (compensation, the bridge, the spur, the permit) enter its own schedule (R1 question 1).
- Criteria: K1's first and second questions (comparable options; material costs and dependencies); K4, a revised input invalidates the run's results and approvals.

CHAPTER 7 · STEP 6

Výsledky: the rule applied, and what could change it

The results answer four questions in order: what the rule chooses, how far the verdict is from flipping, what information would change it, and what the finance, operations and schedule of each option look like. The first question is the recommendation; the rest is evidence about it.

7.1 What goes in

The futures and the base case of step 5; the rule of step 1; the extra computations declared in step 5.

7.2 What comes out

Five views, each a set of tables the report reuses (the exemplar's T1–T6, R2-P06 to R2-P11):

View	Content	Exemplar
Rozhodnutí	the rule table per option (the measure's median and 90 % interval, each condition's value against its threshold, pass or fail), at the rule's rate and the client's other rates; the **decision boundary**: the rule's choice as one decisive assumption moves over its range; delay tables; stress tests; the tornado; break-even values; alternative terms; the correlation check	T1–T3; the flip at 14 / 15 / 16 %; delay 0/2/4/6 months with and without compensation; hiring delay; tornado; break-even prices; post-2035 terms; three correlation settings (R2-P06–P08)
Hodnota informace	two measures, in this order: **the value for the rule** (what knowing an answer in advance and applying the rule again would bring, averaged over the possible answers), then **EVPPI** (the gain in mean saving from knowing one input); then **the enquiry table**: what would be learned, the output, the duration, the cost, and the consequence of waiting	122,5 mil. EUR for the rule at Odrzyca against 4,9 by EVPPI; four enquiries (R2-P09)
Finance	pre- and post-tax NPV at the rates, IRR, payback, investment by item with the probability of staying under the ceiling; incentives by their nature	T4, T5 (R2-P10)
Provoz	cost per unit by category in a full year of operation, base case	T6, 2032 (R2-P11)
Harmonogram	the start-date distribution per option against the deadline (median, 10th–90th percentile), P(on time), and the effect of each month of delay	the start chart and the delay tables (R2-P07); board M4g, new

7.3 Checks

- The rule is applied exactly as written, condition by condition; the pass column is computed, never typed. The report prints the value of every condition against its threshold, pass and fail alike (T1).
- The verdict is given as a boundary, not a point: for the one or two assumptions the verdict rests on, the table of what the rule chooses at each value (R2-P06 § 5.3). A verdict with no boundary table has not been tested.
- A probability is a share of the model's futures under the inputs that produced it; every probability says which assumption it stands on (R2-P04 § 3.3). The report claims no probability for anything the model does not compute (R2-P15).
- The value of information is computed for the rule first and for the mean second; neither is a budget for the enquiry; the enquiry table carries duration and the consequence of waiting (review R7; K5).
- A median is not a mean: the rule's choice of statistic is kept, and the report says what the right tail does to the other (R2-P06 § 5.1).
- The base case and the futures may disagree; the report explains why when they do (R2-P08 § 7.1: break-even in the base case against loss in the median).
- Every extra computation names the one assumption it changes (R2-P15).

7.4 How the exemplar did it

No site passes at 8, 6 or 10 % (T1–T3). The Czech and Polish sites fail only the deadline condition (0,75–0,76 against 0,80 for the three leaders, 0,19–0,21 for Vrbice and Bory); the German sites fail mainly the loss condition (0,14–0,21 against 0,10). The verdict is a boundary: at a compensation chance up to 14 % the rule picks Odrzyca, at 15 % Kamenec, from 16 % nothing; the model uses 20 % (R2-P06 § 5.3). Delay of the decision by 2, 4 or 6 months lowers every P(on time); with compensation ruled out, the rule still picks Odrzyca after two months and nothing after four (R2-P07 § 6.1). Hiring delay is a stress test that adds to decision delay (§ 6.2). The tornado puts the 2026 Hanshin price first (a swing of 134,6–152,7 mil. EUR), the renewal premium second (57,7) and overrun third (34,2–44,5) (R2-P08). Break-even prices lie 1 249–1 308 EUR/t for Czech and Polish sites, 1 413–1 450 for German ones; at the 2025 average of 1 296 EUR/t the median of every site turns negative (§ 7.1). Post-2035 terms are the second lever: at a 30 % premium Wendelstedt would pass (§ 7.2). Three correlation settings move medians by at most 1,4 mil. EUR and probabilities by at most 0,01 (§ 7.3). The value for the rule of the authorities' answer is 115,2–152,1 mil. EUR; EVPPI of the same information is 4,9

mil. EUR, of all inputs 14,2 (R2-P09). Finance: base-case saving 170,7 / 165,9 / 159,9 mil. EUR for the three leaders at 8 %, IRR 18,2–20,5 % in Czechia and Poland, payback 2034–2036; after tax Kamenec and Odrzyca are level (T4). Operations: 1 290–1 311 EUR/t in Czechia and Poland against 1 434–1 463 in Germany in 2032, the difference in staff, maintenance and freight (T6).

7.5 In AIA today

- A number reaches a result only as an admitted claim (`domain/evidence/admission.py`, `application/admitted_results.py`); suppression is decided on the server and a suppressed row prints its reasons, never a number. The pattern holds for model results: a cell the method does not support is not printed.
- The results screens of a research run and their reading in words (`ExecutionSteps.tsx`, `lib/result-reading.ts`): unknown is never ordered, an interval missing means no order.
- The report's print grades and the evidence ledger (`domain/report/evidence.py`), the per-question tables as exports (`domain/report/table_export.py`, RD12).
- The budget line and spend reading as the pattern for "unknown never zero" (`lib/budget-line.ts`, `lib/run-spend.ts`).

7.6 New

- The rule table and the pass computation as a result type with the grades ESTIMATED for everything from the futures and CALCULATED for the base case (`cost-model-study` CP8, CP10, CR7, CR8; board M4 in revision, session C).
- The decision boundary as a declared operation over one input's range, re-running the futures per value (new; the exemplar's § 5.3 and its "hranice rozhodnutí").
- Value of information for the rule, apart from EVPPI (CR6 plans EVPPI only; review R7 asks for the rule's measure first), and the enquiry table with duration, cost and the consequence of waiting (K5).
- The tornado and break-even on the base case (CM2, CM4 cover one-way sensitivity and break-even on one driver); alternative-terms tables and the correlation check (new).
- Finance: IRR with a stated root rule, payback, pre- and post-tax at several rates, investment by item with P(under the ceiling) (CM7, FIN1–FIN3; board M4b).
- Operations: cost per unit by category in a stated year (board M4d); the new *Harmonogram* view, start-date distributions against the deadline (board M4g, new).
- Boards M4c *Účetnictví* and M4e *Trh* are parked for type 2: they have no component in the ChemVitro case.

7.7 ChemVitro acceptance

The answer key's reference model finds CZ3 Kamenec passing at 8 % (median +39,8 mil. EUR, P(beats) 0,82, P(loss > 20) 0,09, P(start) 1,00), CZ2 close behind, PL1 on the edge of the loss limit, nothing at 10 %, and names the renewal premium (12–22 %) as the assumption most worth checking (answer key § 1). The exemplar finds no site passing at any rate and names the compensation chance first and the Hanshin price second. The two differ on one judgement: the reference prices offsets only where a station exceeds the limits; the exemplar adds a 0,2 chance everywhere else in Czechia and Poland, and says that this judgement decides the verdict and has no document behind it (R2-P15).

- The acceptance therefore does not require the exemplar's verdict or the reference's numbers. The key says so itself: "judge a report by whether it found the planted facts and applied the rule honestly, not by whether it hit these numbers."
- It requires: the rule applied as written with every condition's value printed; the verdict given with its boundary on the assumption it rests on; the assumption most worth checking named with its value for the rule; the three traps and the overrun from the comparables visible in the investment and the schedule (CZ1 and PL2 fail the start date; DE3's P(start) falls with the bridge).
- Both the exemplar and the reference agree that the brochure favourite, PL2 Bory, cannot start on time. A study that recommends Bory has read the brochure, not the files.
- Criteria: K3 (the rule's choice kept apart from supplementary scoring), K5 (every enquiry states its effect on the deadline), K6 (the rule table, the summary and the boundary table say the same numbers).

CHAPTER 8 · STEP 7

Sociomapa: the futures map

A Sociomap places the options as points in a landscape: two options lie close when the same observers rate them alike, and the height shows the mean rating. In a type-2 study the observers are the model's futures. The map is evidence about the structure of the options' risks, never about the recommendation.

8.1 What goes in

- The futures of step 5 as a table: one row per future, one column per option, the cell the measure the rule reads (the saving against the baseline in that future).
- The mode contract for futures (board MV3 § 02, mode *Scénáře*): what a row is, what the claim is about, what is measured, what the null means, the resampling, what the stability measures.
- The stability test, declared before the computation: which alternative settings the map is recomputed under and what agreement counts as stable.
- The method pinned: `aia-sociomap-5`, the organization's methodology in force.

8.2 What comes out

- **The futures map** in two views: the 3D terrain and the top view that does not distort distances by perspective (R2-P12 § 11.2). Heights are mean saving; links are drawn only between pairs the method marks reliable; a warm link means co-movement, its width the relation's strength.
- **The relation matrix**, the exemplar's T7: each pair's relation across the futures, from which the map is drawn.
- **The shared-failure table**, T8: when an option is worse than the baseline, the share of those futures in which each other option is worse too. This reads the absolute, uncentred futures; the map reads the centred, relative ones.
- **The stability result:** the maps under each setting, the pairwise agreement of their distance orders, and pass or fail against the declared threshold.
- **The register status:** the map's method at its spec fingerprint, approved or *interní* (`domain/sociomap_register.py`), printed on the map and in the report.

8.3 Checks

- **Mode contract.** A row is one independent draw of the model. The claim is about the model and its assumptions, not about the market. Centring each future on the mean of the options makes the map show relative performance: who does better in the same future, not whether all do well (R2-P04 § 3.3; review R5).
- **Relative apart from absolute.** The question "is option B a backup for A" is answered by the uncentred shared-failure table, not by the map: centring removes the common level, which is exactly what shared failure is made of (R2-P12 § 11.3; board MV3 § 05).
- **Links only where the contract allows.** The exemplar drew links where the `-5` method's pair test, with the false-discovery correction, marked a pair reliable (2 of 28). Board MV3's scenario contract says a null makes no sense when the model creates the relations, and asks instead for stability across seeds and assumption changes, with no p-values. Which of the two applies to futures is an open question for the methodologist (chapter 13); until it is answered, the report prints which rule it drew by.
- **Determinedness.** Split-half agreement of two halves of the futures against the method's threshold says the layout is not an artefact of the draw (0,91 against 0,59 in the exemplar). It does not say the layout is stable under the assumptions.
- **Stability, declared in advance.** The map is recomputed under the alternative settings written down in step 4 (the exemplar: no correlations, the judgement's correlations, strong correlations) and counts as stable only if every pair of versions agrees in the order of the inter-option distances at or above the threshold (0,8 in the exemplar, Appendix B). Failure is printed, not softened.
- **What the map never claims** (board MV3 § 06): causation; the behaviour of the market or the firm; that a relation is observed rather than a property of the model; that a distance is an effect size in any unit; that the terrain between points is an estimate; validity borrowed from team-communication Sociomaps, which do not cover this use.
- **Internal until approved.** Maps whose observers are futures or properties are an estimate awaiting the methodologist's approval; they print *odhad (metoda čeká na schválení metodikem)* and carry the internal label (R2-P04 § 3.2). The Sociomap of people's ratings is approved since 21. 10. 2026; this is a different mode.

8.4 How the exemplar did it

Eight sites, 20 000 futures as observers, each future centred on the mean of the eight (R2-P04 § 3.3). The map divides the sites by country: the German sites lie together in a valley (74,2 mil. EUR mean saving at Eschfeld), the Czech and Polish sites on a ridge (147,6 at Odrzyca). Within countries the relations are strong (Lindenau–Eschfeld 0,71, Wendelstedt–Eschfeld 0,70; Lipová–Kamenec 0,55; Odrzyca–Bory 0,48); between countries weak and negative, −0,07 to −0,27, the sign partly a consequence of centring (T7). Two of 28 pairs were marked reliable. Split-half agreement 0,91 against a threshold of 0,59: determined. The stability test failed: the judgement version agrees with the no-correlation version at 0,95 and with the strong version at 0,84, but those two agree only at 0,75, under 0,8; the relations in the matrix are steadier than the drawn positions (0,966, 0,899, 0,826). Without correlations seven pairs are reliable and the Czech sites merge; with strong correlations no pair is and Vrbice moves away. The report therefore reads from the map only the heights and the split into a German group and a Czech–Polish side, which appears in every version, and nothing about fine positions, causes or probabilities (R2-P12 § 11.2). For the backup question it uses T8: the best backup fails together with the chosen site in 75–79 % of its bad futures, the German sites in 97–100 %; no site is a backup of another (§ 11.1, 11.3). The version without correlations did not converge in the program's log, one more reason to read positions carefully (R2-P15).

8.5 In AIA today

- The `aia-sociomap-5` engine, signed off by the methodologist on 21. 10. 2026 and what a new run pins: `domain/sociomap/signed_off_inputs.py` (per-person centring, the baseline, pair floors, Benjamini–Hochberg), `signed_off_layout.py`, `signed_off_credibility.py` (split-half agreement, map uncertainty), `signed_off_respondents.py`, `signed_off_segments.py`; `engine_v2.py` `compute_object_map` and the contract-3 artefact.
- The register: which method at which spec fingerprint is approved; anything absent is `INTERNAL_ONLY`, never presumed (`domain/sociomap_register.py`, ADR 0038).
- Map snapshots drawn on the server from the stored map at the report's camera and approved as artefacts (`application/map_snapshots.py`, RD4); the map chapters of the report (`application/report_maps.py`, `infrastructure/report_docx/object_map_figure.py`).
- The canvas's own proof that the engine takes a futures table as rows: `prototypes/obchodni-model/tools/engine_maps.py` ran `compute_object_map` on 2 000 futures and drew the maps with the report's renderer.
- The web's map: `ObjectMapPanel`, `lib/object-map-signed-off.ts`, `lib/sociomap-display.ts`, display only.

8.6 New

- A Sociomap stage that takes a model run's futures as rows (`cost-model-study` CS1–CS5 plan a properties matrix and the WIND height; the futures map is a different input and needs its own chunk; board MV3 [in revision, session D]).
- The mode contract as data, per mode, printed with the map (board MV3 § 02; review R5).
- The stability test as a declared operation: alternative settings from step 4, the agreement measure, the threshold, the verdict (new).
- The relation matrix T7 and the shared-failure table T8 as result tables of the run (new).
- A register entry for the futures mode (chapter 13).
- The weighting-scenario map, a second map beside this one, on board MV9 (new; chapter 11).

8.7 ChemVitro acceptance

- The eight sites placed by whether they do well in the same futures; links only where the declared rule allows; the view chosen and the reason stated (`STUDIES.md` § 2).
- The map labelled internal until Metodika approves the method.
- The relation matrix and the shared-failure table printed; the backup question answered from the table.
- The stability test declared before the computation, and its result printed whether it passed or failed.
- Criteria: K3 (the map is not part of the recommendation), K6 (the matrix, the text and the figure agree).

CHAPTER 9 · STEPS 8 AND 9

Osnova a Zpráva: the outline, then the report

The outline comes before the text. It lists every claim the report will make in the order of the argument, each with its source or method and why it matters to the decision. The report then says only what the outline holds.

9.1 What goes in

- The results, the map, the input table, the findings table, the public inputs and the decision record: every artefact of steps 1–7, by id.
- The part structure of a type-2 report and the grade and source conventions (below).
- The validity review of chapter 13, before approval.

9.2 What comes out

The outline (*Osnova zprávy*, board MV4)

Five parts in the order of the argument (review R10): 1 the decision and what is asked of the board; 2 why the preferred option is acceptable, or why none is; 3 what could change that conclusion; 4 implementation conditions, owners and dates; 5 supporting evidence and technical appendices. Each point carries its source or method, its grade attributes, and one line "why this matters" (*proč na tom záleží*). A judgement is a typed judgement record; a synthesis from sources is an inference record with its evidence (review I5). Stops (*zastávky*) close parts for review; a closed stop may go to the client's decision-makers for comment.

The report (*Zpráva*, board M6)

The exemplar's part structure, which the MVP reproduces:

Part	Content	Rule
R2-P01	*Shrnutí*: *Doporučení* · *Důvod* · *Jistota* · *Další rozhodnutí*, the rule table, *Stav podkladů*	one page plus the state box; the rule table is T1 without the intervals; the state box names the gaps that could change the result and the public inputs not yet verified
R2-P02	*Úvod*: the situation, the options, the rule	the rule printed in full with its date and the interpretation record
R2-P03	*Přehled literatury*	each work with "what follows for this report"
R2-P04	*Metodika*: about AIA, the Sociomap, the model in words, the data, the grades	the grade legend is printed in every report; the Sociomap paragraph says which modes are approved
R2-P05	*Předpoklady*: the assumptions the verdict rests on	the ones without a document first
R2-P06–P11	the result chapters: decision, schedule, levers, information, finance, operations	each chapter's first sentence is its finding; each number carries its grade; sources numbered at the chapter's end
R2-P12	the map chapter	relative and absolute readings, the stability result, what the map may and may not say
R2-P13	*Diskuse*	what the results mean together, and what the model does not resolve
R2-P14	*Doporučení a provedení*	a dated plan: when, what, who, output; the enquiries with their owners; what to watch
R2-P15	*Meze studie*	every judgement without a document, every provisional input, every simplification, every method awaiting approval; "each change computed separately"
R2-P16–P17	*Literatura, Seznam zkratek*	—
R2-P18	*Přílohy* A–D and tables T1–T9	A the document register; B the method in detail and the findings P1; C data checks and computation (seed, N); D the tables, generated by the model from the same run

Every number carries one of the grades of chapter 12 in small type and a numbered source at the chapter's end; the summary's recommendation status says *předběžné* where the recommendation rests on a method outside the register (chapter 12).

9.3 Checks

- No point without a source or a method; a point whose source was not captured does not enter the report (MV0 M-9; board MV4's point 3.4).
- The report says only what the closed outline holds. A sentence that compares in words with no number behind it is marked (RD7's sentence check, already built).
- Every number is a reference to a stored result: the same run, the same artefact id. Tables are generated by the model from the run, so they agree with the text by construction (R2-P18 Appendix D; K6).
- The required disclosures are in every report: the recommendation's status and qualifications, each number's source and method, what is outside the model, who stands behind the assumptions, the method versions (board MV4's *Povinná zveřejnění*).

- The validity review is recorded before approval, and the approval records the exact version and the unresolved qualifications (review R1, R9; chapter 13).
- A stop closed with the client's people records their agreement (ADR 0046 W26).
- No respondent, person or e-mail is named beyond a role; the report is Czech in the MVP (ADR 0046 W28).

9.4 How the exemplar did it

The report's first page is the summary in four paragraphs and the rule table; the second is the state of the inputs. Every chapter's title is its finding ("No site yet passes the rule", "The cheapest sites do not meet the deadline certainty", "The saving rests on today's price from Hanshin"). Every number carries its grade in small type, and every chapter ends with *Zdroje a stupně*, numbered. The part ids R2-P01 to R2-P18 let the executive comment part by part. The limits chapter admits what the report cannot claim: the compensation chance is a judgement, public inputs are provisional, the map is not stable, the methods of the futures await approval, each change was computed separately (R2-P15). The implementation plan is a dated table with owners and outputs (R2-P14). The tables T1–T9 are generated from the same run as the text (R2-P18).

9.5 In AIA today

- The report as data: the document model with sections and blocks, numbers only as refs (`domain/report/model.py`, `document_model.py`); the evidence ledger built from admitted claims with print grades (`evidence.py`); the section-source manifest naming each section's sources (`sources.py`); printed numbers as locked content controls with one tag grammar (`control_tags.py`, `controls.py`).
- The DOCX renderer with deterministic bytes, the Word style sheet from the print tokens, tables with repeating headers and suppressed rows counted, evidence marks as vector images (`infrastructure/report_docx/`); the PDF beside it (`infrastructure/report_pdf.py`).
- Composition: the internal report, the client report, the report with literature, their chapters from verified sources (`application/report.py`, `report_chapters.py`, `report_tables.py`, `report_design.py`, `report_method.py`, `client_report.py`, `contextual_report.py`).
- Approval once per version with the approver recorded and self-approval rules (`application/report_approval.py`, `artifact_repository.approval_record`); the client gate's checks as warnings (`domain/report/client_gate.py`, RD0); the sentence check (`sentence_check.py`, RD7); the report editor with saved versions and their changes against what code wrote (`report_editor*.py`, `saved_changes.py`).
- The stored document read on screen with each number a chip to its claim (`ReportDocumentView`, `lib/workbench-artifacts/report.ts`).

9.6 New

- The outline stage with points, sources, "why it matters", judgement and inference records and stops (`research-findings-stage`; board MV4 in revision, session E).
- Composition per situation from chapter kinds (ADR 0045) with the type-2 part structure R2-P01…P18 as a template, and the argument order (review R10).
- The grades of chapter 12 as print grades beside today's five (`cost-model-study` CP8 adds CALCULATED, CP10 ESTIMATED; the client-reported, judgement, public-source and legal grades are new), the four attributes per number, and the numbered sources per chapter (new).
- Tables generated from the run into the report (the exemplar's T1–T9), the state box, the limits chapter composed from the input table's states and the run's declared simplifications (new).
- The validity review record and the approval that records version and qualifications (board MV8, chapter 13).

9.7 ChemVitro acceptance

- Every number carries its grade and source: file and row or page, or the quoted public source (`CASE.md` § 11).
- An estimate from a method that is not approved is labelled *odhad (metoda čeká na schválení metodikem)*.
- The report is Czech, as PDF or DOCX; the seven parts of `STUDIES.md` "What every report contains" are present.
- No respondent named or quoted by id.
- Criteria: K6 is checked here, by code and by a person: no contradiction between narrative, tables and figures, percentile and probability wording included. The canvas's own K6 finding (SI3 against KM4) shows the check is needed.

CHAPTER 10

Built vs designed

No type-2 code is on `develop`. The exemplar's model ran outside the product. What AIA has is the ground the pivot builds on: the shell, the working copy, client sources, Deep Research, the Sociomap engine and its register, the report document model and its renderer, approvals and audit. This chapter says which is which.

10.1 The state on 8 November 2026

Step	Built on `develop` (reused)	Designed (board; plan chunk)	Neither yet
Zadání	working copy, revisions, edit ops; Brief stage; the Lead; approval ledger; material classes	Z1–Z7; `cost-model-study` CP3 (study kind), CR7 (the rule)	interpretation records; perspectives chosen per case
Data	client sources with readers and locators; dataset preview with personal columns; intake sheet; evidence policy (ADR 0027/0037)	M1, MV1; `client-data-in-analysis` 2, 2a, 3, 3a, 3b, 10; `cost-model-study` CR1	findings table with dispositions; per-column unit, currency, price year; document register
Rešerše	Deep Research with grounding and the sealed bundle; RetrievalGate; Wikipedia, Brave, Common Crawl; verified sources	MV2, MV2b; CP1's `public_source` provenance; MV0 names `public-data-routes` (no plan file)	typed public inputs with state; verification at source as a dated act; the first-page state line
Předpoklady	methodology as data (versions, activation, bounds); ledger and audit; the simulation core's typed, validated inputs as a pattern	M2; CP1, CR2, CR3	the input table with guarantor and state; empirical joint draw from comparables; scope check of a supplied estimate; mixtures for expert disagreement
Model	worker, workflows, artefacts with SHA-256, pipeline fingerprints; methodology pins on a run; seeded determinism	M3, M4; CP1–CP7, CM1–CM7, CR3, CR4; `quantitative-families` FIN1–FIN3	a type-2 model type with a shared timeline, a bought baseline and residual value; the stopping rule; the extra computations as declared operations
Výsledky	admitted claims and suppression; results reading (unknown never ordered); print grades; table exports	M4, M4b, M4d, new M4g; CP8, CP10, CR6–CR8, CM2, CM4, CM7	decision boundary; value for the rule and the enquiry table; alternative-terms tables; the correlation check; start-date distributions
Sociomapa	`aia-sociomap-5` engine; the register; map snapshots; report map chapters; the canvas tool that maps futures with the engine	MV3, SI1–SI4, new MV9; CS1–CS5 (a different input)	futures as rows in a product stage; mode contract as data; the stability test; T7 and T8; the weighting-scenario map; register entries
Osnova, Zpráva	document model, evidence ledger, control tags, source manifest; DOCX and PDF renderers; composition; approval; client gate; sentence check; report editor	MV4, M6, MV8, MV5; `research-findings-stage`; `report-delivery` RD3, RD4, RD9; ADR 0045	the type-2 part template and argument order; the seven grades as print grades and the four attributes; numbered sources per chapter; generated tables; the validity review record

10.2 The plans that carry the build

From `python tools/progress.py` on 8 November: `cost-model-study` is planned, 0 of 32 chunks; `client-data-in-analysis` in progress, 1 of 15 (8a built); `report-delivery` in progress, 7 of 16; `quantitative-families` planned, 0 chunks begun. ADR 0044 (quantitative models as a study kind) and ADR 0045 (reports composed per situation) are Proposed. ADR 0046 (the pivot) is Proposed on branch `chore/pivot-business-decisions`. The MVP scope (`MVP-SCOPE.md`, session A) maps each step to the chunk that builds it; this chapter must agree with it, and the coordinator folds in the scope when it lands.

10.3 Three honest statements

- **The exemplar was not produced by AIA's product.** Its author computed the model with his own program and wrote the report by hand to AIA's conventions. The report's sentence "every number was computed by a verifiable program" is true of that program. The MVP's job is to make it true of AIA.
- **The canvas is design.** The thirty boards show the screens and the method; their numbers are fictional, and the Velis case's numbers are illustrative, not validation evidence (board MV5). The boards hold the method; the plan chunks hold the build.
- **The methods of the futures are not registered.** Monte Carlo with PERT and correlations, the decision boundary, both measures of the value of information, the futures map and the weighting-scenario map await the methodologist's entries (chapter 13). Until then their results print *odhad (metoda čeká na schválení metodikem)*, and a recommendation that rests on them is *předběžné* under R9, which itself waits for the owner.

CHAPTER 11

Perspectives, components and the two maps

The owner asked for every business perspective: financial, strategic, operational, marketing, logistics, "just everything". A perspective is not a view of the results. It is a set of components of the decision, each a measure per option with a source and a direction. The components feed one new map, the weighting-scenario map, which sits beside the futures map and is never merged with it.

11.1 Perspectives and components

A **component** is one measure per option with: a **source** (a model result, a client datum, a cited finding, or a named expert's score); a **direction** (higher is better, or lower is); a **value scale** with declared extremes, so that the component can be read on 0–1; and a **state** per option (known, unknown, not applicable, evidenced absent or present; review R2). A **perspective** groups components. A perspective with no component for a case stays empty and says so.

The seven perspectives, with ChemVitro's components read from the exemplar. Every value below is the exemplar's or the case pack's; a component the case does not have is marked so.

Perspective	Components in the ChemVitro case	Source	Direction and state
finanční	median saving; P(beats buying); P(loss > 20); investment in the base case and P(≤ 175); IRR; payback; after-tax NPV	model: T1, T4, T5	saving, probabilities of beating, IRR higher; loss, investment, payback lower; known for every site
strategická	supplier dependence (binding offer only mid-2029; renewal at +12–22 % and a 30 000 t minimum); reversibility (a plant is built once, no pilot); post-2035 terms as a lever; market position	client documents (the supplier's letter), `CASE.md` § 8; model § 7.2	dependence and irreversibility lower; post-2035 terms known only as scenarios; market position **not applicable**: the plant supplies ChemVitro alone
provozní	cost per tonne in 2032 by category; staffing 95 FTE; ramp-up 60–90 %; time to fill an operator post (49–118 days); a large employer hiring 380 nearby	model T6; `trh_prace_regiony.xlsx`	cost and hiring time lower; staffing identical by design; the Kamenec hiring is a risk, not a cost
marketingová / obchodní	customer promise, delivery time, demand growth	—	**thin for ChemVitro**: demand is the client's own consumption (34 000 t, +1–3 %); the perspective is central for Velis and stays empty here, and the report says so
logistická	product freight to Brno and Pardubice (20–38 EUR/t from CZ and PL, 53–58 from DE); rail access (the PL1 spur, 3,4 km); road access (the DE3 bridge, 16 t); distance	model T6; `vyjadreni_doprava.pdf` p. 6–7; `doprava_nabidky.csv`	freight and distance lower; the spur and the bridge are costs and months in the model and gates until evidenced
prostředí a povolování	permitting 8–11 months plus extensions; compensation chance (0,8 / 0,2 / 0); wastewater pre-treatment (CZ2, DE2); emissions in the EU ETS (0,62 t CO_2 per tonne); station values against limits; the 2030 limits of (EU) 2024/2881	model T9; `memo_povolovani_ovzdusi.pdf`; `ovzdusi_stanice.csv`; `vyjadreni_cov.pdf`	months, chance and cost lower; the compensation chance is a judgement with no document, so its state is **unknown** at the three leading sites
lidé	operator wage and the wage mix (× 1,10–1,50); employer contributions (CZ 34–39 %, DE 20–23 %, PL 20–24 %); unemployment; hiring time	`trh_prace_regiony.xlsx`; Rešerše (provisional)	wage and contributions lower; contributions are *prozatímní*

A component whose mechanism the model already holds is not scored a second time (the review's "no double counting"; the canvas's version 1 counted one option's fixed price three times). For ChemVitro, most components above are already inside the saving; the perspectives make them visible per option, so that the board can weigh them, and that is what the second map is for.

11.2 The futures map

Chapter 8 describes it. In one line each: rows are futures; the claim is about the model; the cell is the rule's measure; centring per future makes the reading relative; the uncentred shared-failure table reads the absolute; links only where the mode contract allows; a stability test over settings declared in advance; relative apart from absolute; internal until the futures mode is registered.

11.3 The weighting-scenario map (*Mapa scénářů vah*, board MV9)

The owner's request: "allocate different weights to different components … whether logistics is a 0,3 and operations management is 0,5 … run the scenarios and display the scenarios on the Sociomaps."

Rows	**Scenarios**: named weight vectors over the components or the perspectives, composed by the researcher or the client's people. Example names, to be replaced by the case's own: *FR: finance 0,5 · provoz 0,3 · logistika 0,2*; *EHS: povolování 0,5 · prostředí 0,3 · finance 0,2*; *Rovné váhy*. Around each named scenario a **fan**: perturbed weight vectors within a declared distance, renormalised, so that a scenario is a region, not a point.
Objects	The options (the eight sites; the baseline where it has components).
Cell	The option's weighted score under that weight vector: each component read on its 0–1 value scale in its direction, multiplied by its weight, summed.
Height	The weighted score under the scenario chosen for display.
Distance	Options close together score alike across the tested weightings: they win or lose under the same preferences.
Statistics	None of the engine's pair statistics are read: the rows are settings we chose, not a sample (board MV3's scenario contract; README "Known wrong"). Instead, **shares of tested settings**: for each option, the share of the fan's settings in which it ranks first (review R4). A share is a share of tested settings, never a probability.

The rules the map keeps

- **The client's rule decides.** The map is **preference information**. It goes to the report's part 3, *Co by závěr změnilo*, and to the appendix, never into the recommendation (review R3; K3).
- **Gates are never compensated.** The rule's four conditions and any non-compensable requirement are gates: an option that fails one cannot rank first under any weighting; an option whose gate is unknown "cannot be recommended until evidenced" (review R2). The map shows gated options apart or marks them.
- **A missing component keeps a range, never a score.** An option with an unknown component is scored as an interval over the component's possible values; the share counts the settings in which it ranks first under its worst value and under its best, and the map prints the coverage of each option's weights.
- **Weights come from the decision-maker's trade-offs** (swing weighting), approved with the value scales before any score is read; other stakeholders' weights are a sensitivity test (review R3).
- **Two maps, never one.** The weighting map and the futures map answer different questions (preferences; shared risks in the model). They are shown side by side with their own contracts and are not merged, as the respondent and model maps are not (ADR 0046).
- **The engine is reused, the statistics are not.** The layout and terrain of `aia-sociomap-5` draw the map; the split-half and the pair tests are not printed for this mode. The map is internal until the indicator method v2 (states, gates, swing weights, value scales) has a register entry (chapter 13).

What the exemplar shows for it

The exemplar has no weighting map; it has the material for one. Its components per site are in T1, T4, T5, T6 and T9, and its text already weighs them in words: the Czech and Polish sites win on finance and operations, Wendelstedt on the schedule, the German sites on permitting certainty, and none on all. A weighting map would show that the three leaders win under every finance-heavy weighting, that Wendelstedt wins only when the deadline condition is read as a weight rather than a gate, and that the gate is what keeps it out. That is exactly the kind of statement the map may make: what the preferences would favour, not what the rule chooses.

> **In revision.** Session D builds MV9 on the real engine with the perspectives above; session A defines the perspectives for every session in `MVP-SCOPE.md`. The names, the fan distance and the value scales in this chapter are placeholders until those land, and this chapter will be aligned to them.

CHAPTER 12

Grades, sources and the release gate

Every number in a type-2 report carries a grade in small type and a numbered source at the end of its chapter. The grade says how the number came to be; the source says where. Behind the one printed grade stand four attributes.

12.1 The exemplar's seven grades

Grade	Meaning (R2-P04 § 3.5)	Exemplar instance
z dat klienta	a number read from a client's file or document	contract end 30. 6. 2030; 36 comparable plants
výpočet	a number the program computed from stated inputs in one computation: the base case, break-even, finance, operations	base-case saving 170,7 mil. EUR; T4–T6
odhad	a range derived from client data, for instance from comparables; and every number from the futures, which carries the full label *odhad (metoda čeká na schválení metodikem)* while the method is unregistered	overrun median 17 %; every median, interval, probability, value of information, boundary, delay table and the futures map
úsudek	the author's assumption where the client has no answer; also proposed dates	the 20 % compensation chance; the correlations; the plan's dates
veřejný zdroj	a figure from public research, provisional until verified at the source	allowance price 89,10 EUR/t; exchange rates
právní předpis	a provision, likewise provisional until verified	tax rates, employer contributions, the directive's limits
průzkum	answers from a survey; type 2 does not use it	—

12.2 The four attributes behind a grade

Version 2 of this methodology, following the review's I3, split a number's label into four attributes. The printed grade is a projection of them; the outline and the screens hold all four (boards MV4 and M6; the other boards' chips still show one grade, README "Known wrong").

Attribute	Values	Projection to the printed grade
Source	*data klienta · veřejný zdroj · právní předpis · úsudek · model*	the source alone gives *z dat klienta*, *veřejný zdroj*, *právní předpis*, *úsudek*
Method	*přímo · výpočet · odhad · modelováno*	a model source prints its method: *výpočet* or *odhad*; *modelováno* is the panel respondents' grade (ADR 0024) and does not occur in type 2
Uncertainty	a range, an interval, a probability's basis, or a certainty	printed beside the number, not in the grade: "90 % interval −23,8 to 359,1"
Validation	*metoda v registru · čeká na metodika*; for a public input *ověřeno · prozatímní*	adds the suffix *(metoda čeká na schválení metodikem)* to *odhad*, and the state *prozatímní* to a public input

In the code the print grades are five today, `PrintGrade` in `domain/report/evidence.py`: measured, calibrated, modelled, holdout-pending, unknown, strongest first, a claim printing the weakest grade of the fields behind it, a missing grade printing "?". The plan adds CALCULATED (*výpočet*) and ESTIMATED (*odhad*) with the register's status (`cost-model-study` CP8, CP10), and ADR 0027 the client-reported origin. The judgement, public-source and legal grades and the provisional state are new. The weakest-grade rule carries over: a number computed from a judgement input prints *výpočet* but its sensitivity to that input is shown; a number from the futures prints *odhad* whatever its inputs.

12.3 Sources, numbered per chapter

- Each chapter ends with *Zdroje a stupně*: a numbered list; each printed number cites a list entry by its number (R2-P02: "z dat klienta 1").
- An entry names the document by its cited name and page or paragraph ("Zápis zadání s klientem §6", "Dopis dodavatele, s. 1"), the file for data, the table for a computed value ("výpočet: tabulka T5 v příloze"), or the method's status for an estimate ("odhad (metoda čeká na schválení metodikem)").
- An entry may carry two grades when a number rests on both ("výpočet; z dat klienta: Srovnatelné závody"); the weaker is printed at the number.
- The document register (Appendix A) maps every cited name to its file, so a source is checkable with one open.

- In AIA, the section-source manifest (`domain/report/sources.py`) already records each section's sources by artefact id and SHA-256; the numbered list is its printed form.

12.4 The provisional state

Prozatímní is a validation status, not a grade. It marks a public input not yet verified at the source, and it is printed in three places: the input table's state column (T9), the sentence that uses the input ("dlouhodobý vývoj cen energie (prozatímní údaj)"), and the first-page state box with the count (16 of 54). It is removed only by a dated verification by a named person. The report's running head carries it while any decisive input is provisional: the exemplar prints *Pracovní verze · prozatímní* on every page.

12.5 The release gate (R9, proposed)

> **Waits for the owner.** R9 changes two earlier decisions: W27 (an estimate whose method is not registered still reaches the client report, labelled, with a warning, never refused) and report-delivery RD0 of 22 October ("warnings for all of it"). Boards MV0 (M-5), MV5 (§ 04) and M6 show it as proposed.

As proposed, for a business-decision report:

- Every decision-critical claim is checked twice: support in its source (code) and correct interpretation (a person who is not the claim's author).
- A recommendation that rests on a method outside the register is *předběžné doporučení* and is never released as approved investment advice. The exemplar's recommendation rests on the futures, so it is provisional in this sense, and its summary would say so under R9.
- Exploratory material enters the report only with explicit limits ("exploratory; does not support the recommendation"). The weighting-scenario map is such material.
- Approval records the exact version and the unresolved qualifications; "approved" means ready in substance, not merely checked.
- The Deep Research thresholds of M-7 (citation precision ≥ 90 %, unsupported claims < 10 %) are one check, not the release criterion.

What exists today: approval once per stored version with the approver and self-approval rules recorded (`application/report_approval.py`); the client gate's checks as warnings recorded on the approval's ledger row (`domain/report/client_gate.py`, RD0); the sentence check (RD7). What R9 adds: the recommendation's status as a field of the approval, the qualifications list, and the second, human reading of each decision-critical claim.

CHAPTER 13

Governance

Reproducibility proves the code computes what is written. It does not prove the model is the right one for the decision. That needs a review by someone who did not accept the inputs, a register that says which methods are approved, and a record of what is still open and whose word closes it.

13.1 The validity review (R1)

Before a recommendation is approved, an independent reviewer answers five questions; the researcher who accepted the inputs answers to them, never reviews them (board MV8 § 01; review R1).

#	Question	In the ChemVitro exemplar
1	Is every option represented comparably?	the same plant, horizon, rate and timeline; site-specific months and costs from the site's documents (R2-P02, Appendix B)
2	Are the material costs, benefits, constraints and implementation dependencies in the model?	the EPC exclusions priced; outside the model: transfer prices, hiring time in the schedule, raw-material freight, the annual step (R2-P15)
3	Could an excluded item reverse the recommendation?	raw-material freight would lower savings by 9–14 mil. EUR without changing the verdict (P1); the 2030 air limits could (R2-P01)
4	Do extreme inputs give sensible results?	the decision boundary 0–30 %, delays to 6 months, premium 0 and 30 %, three correlation settings (Appendix B); not run as named extreme-input tests
5	Has an independent reviewer challenged the structure and the decisive assumptions?	not recorded in the exemplar; the MVP adds the record (board MV8)

With the questions: four extreme-input tests with the expected behaviour written before they run (MV8 § 02), the statement that validity is not reproducibility (§ 03), and the six acceptance criteria K1–K6 with their state (§ 04). The review's record is part of the approval; its open items keep the recommendation *předběžné*.

13.2 Register entries the exemplar waits on

The Sociomap register (`domain/sociomap_register.py`) and Metodika's versions (`domain/methodology/`, ADR 0023) hold what is approved. A type-2 study needs entries that do not exist. Until each exists, its results print *odhad (metoda čeká na schválení metodikem)*.

Method	What the entry must fix	Where it is used
Monte Carlo over the input table: PERT, declared correlations (Iman–Conover), empirical joint draws from comparables, yes/no events, mixtures	the default distribution by source; the correlation method; the stopping rule and its margin; the seed convention	every median, interval and probability (T1–T3, T5)
The decision boundary	the grid over an input's range; recomputation of the futures per value; how the flip is reported	R2-P06 § 5.3
Value of information for the rule, and EVPPI	the partition of futures by the answer; the rule applied per partition; EVPPI by Strong and Oakley (2013); the order of the two measures; the enquiry table's columns	R2-P09
The futures map (observers = futures) under `aia-sociomap-5`	the mode contract: which pair statistic, if any; the determinedness test; the stability test's settings, agreement measure and threshold; the relative and absolute readings	R2-P12, T7, T8
The weighting-scenario map (indicator method v2)	states, gates, swing weights, value scales, the fan's distance, shares of tested settings	board MV9
Estimates from comparables	the filter, the statistic, the minimum sample, how a joint draw is recorded	overrun and delay, pre-treatment, spur costs (T9)

13.3 The methodologist's open questions

ADR 0046 lists ten (§ Open: multiple testing on maps; levels or changes; effective sample size; resampling for 24–36 rows; the surrogate null; the relation statistic; the synthetic suite's thresholds; the discount rate in probabilistic rules; PERT inputs; forecast evaluation), with board MV5's recommendation for each. The exemplar adds five:

1. **Which contract governs the futures map.** The exemplar drew links by the `-5` pair test with the false-discovery correction and tested determinedness by split-half; board MV3's scenario contract says no null applies when the model creates the relations and asks for stability across seeds and settings instead. One of the two, or a stated combination.
2. **The stability test.** The agreement measure over the 28 distances and the 0,8 threshold are the author's; the three correlation settings ("strong" = 0,95 / 0,85 / 0,4) are judgements.
3. **PERT from three judged points** for the 22 inputs the files do not hold, and when a mixture or a uniform is the better default.
4. **The empirical joint draw** of overrun and delay from 36 plants: whether a sample of that size supports it, and how a 91 kt outlier is treated.
5. **Value for the rule** as a measure: its definition when the rule's choice is "nothing", and how it combines across several enquiries (the exemplar's 152,1 for all three authorities).

13.4 The owner's open decisions

Id	Decision	Recommendation on the canvas
R9 / Q7	The release gate: a recommendation on an unregistered method stays *předběžné*; changes W27 and RD0 of 22 October	accept, for the business study's recommendation only; warnings elsewhere stay (MV0 § 06)
Q1	The four attributes of a number as the vocabulary of the outline and the report	the legend on M6 and the labels on MV4
Q2	Enforce data classes before a client's data flows into a model	yes for Class A in staging and production
Q4	The SOMECS name and licence	until checked, the map is *Sociomapa systémů*
—	Whether the type-2 model is a second model type or an extension of *scénáře nákladů* (chapter 6)	for sessions A and C to propose

Settled by the brief of 8 November: the case for the review's step 5 is ChemVitro, and the MVP's spine is type 2 end to end on it. Settled by the exemplar's client, not the owner: the interpretation of the investment condition (the CFO confirms).

13.5 Change propagation (K4)

A revised input is a new revision of the input table. Every result, table, map and approval that read the old revision is marked stale until someone reviews it; the report's version carries the revision it was composed from. In AIA today the pipeline's fingerprints and the impact rule (`domain/pipeline.py`, the research impact route) do this for a research design; the model platform reuses the rule (`cost-model-study` CP5: inputs frozen by artefact id and SHA-256).

APPENDIX A

Glossary

Term	Meaning in this methodology
Base case (*základní případ*)	one computation with every input at its likely value; compares options and feeds sensitivity; not an estimate of the saving
Baseline	the option every other is compared with; for ChemVitro, keep buying
Component	one measure per option with a source, a direction, a value scale and a state; perspectives group components
Data contract (*datová smlouva*)	which columns of a client file are used and how; units, currency, price year, personal columns, period; accepted by a person
Decision boundary (*hranice rozhodnutí*)	what the rule chooses as one assumption moves over its range, the futures recomputed for each value
Disposition	what was done with a data finding: used; corrected with the reason; left out; taken as an assumption with a range
EVPPI	expected value of partial perfect information: the gain in mean outcome from knowing one input before deciding (Strong and Oakley 2013)
Fan	the perturbed weight vectors around a named weighting scenario
Futures (*budoucnosti*)	the seeded draws of all inputs at once; probabilities are shares of them
Futures map (*mapa budoucností*)	a Sociomap whose observers are the futures and whose objects are the options; centred per future
Gate	a non-compensable requirement; fail or unknown blocks the recommendation
Grade (*stupeň*)	the small-type label on a printed number: how it came to be; chapter 12
Guarantor (*stojí za ním*)	the person who stands behind an input's range
Interpretation record	a reading of a rule's condition that the rule does not fix, with the alternative, its effect and who confirms
Mode contract	per Sociomap mode: what a row is, what the claim is about, what is measured, what the null means, the resampling, what the stability measures
Outline (*Osnova zprávy*)	the report's claims in the order of the argument, each with its source and why it matters, before any text
Perspective	a set of components: financial, strategic, operational, marketing/commercial, logistics, environment and permitting, people
Provisional (*prozatímní*)	a public input not yet verified at the source
Register (*registr metod*)	the methods the methodologist has approved, each at its version and fingerprint
Rule (*pravidlo*)	the client's decision rule, written before any result: the measure, the choice, the conditions with thresholds
Shared failure	when one option is worse than the baseline, the share of those futures in which another is worse too (T8)
Share of tested settings	the share of a fan's weight vectors under which an option ranks first; not a probability
Stability test	the map recomputed under settings declared in advance; stable only if the distance orders agree above the threshold
Stopping rule	how many futures to run: until each condition's verdict is known within a margin, or a declared N
Value for the rule (*hodnota pro pravidlo*)	what knowing an answer in advance and applying the rule again would bring, averaged over the possible answers
Weighting-scenario map (*mapa scénářů vah*)	a Sociomap whose rows are weight vectors and whose objects are the options; height is the weighted score; preference information

APPENDIX B

The exemplar in numbers

Condensed from the exemplar's tables (R2-P18). Every value is the exemplar's; the grades are its grades. NPV 2027–2044, mil. EUR, pre-tax, 8 % unless stated. odhad marks a number from the 20 000 futures (method awaiting approval); výpočet a base-case computation.

B.1 T1 · The rule at 8 %

Site	Median saving	90 % interval	P(beats) ≥ 0,75	P(loss > 20) ≤ 0,10	P(start by 1. 7. 2030) ≥ 0,80	Investment ≤ 175	Passes
Vrbice (CZ1)	110,8	−40,7 … 331,5	0,88 ✔	0,08 ✔	0,21 ✘	160,6 ✔	no
Lipová (CZ2)	126,0	−30,2 … 349,4	0,91 ✔	0,06 ✔	0,76 ✘	158,6 ✔	no
Kamenec (CZ3)	132,4	−23,5 … 356,9	0,92 ✔	0,05 ✔	0,76 ✘	152,3 ✔	no
Wendelstedt (DE1)	82,3	−74,7 … 308,1	0,80 ✔	0,15 ✘	0,94 ✔	171,8 ✔	no
Lindenau (DE2)	86,5	−68,8 … 311,7	0,81 ✔	0,14 ✘	0,77 ✘	170,8 ✔	no
Eschfeld (DE3)	61,2	−94,4 … 285,7	0,73 ✘	0,21 ✘	0,67 ✘	175,5 ✘	no
Odrzyca (PL1)	134,7	−23,8 … 359,1	0,92 ✔	0,05 ✔	0,75 ✘	148,9 ✔	no
Bory (PL2)	123,2	−30,9 … 344,4	0,91 ✔	0,06 ✔	0,19 ✘	143,9 ✔	no

Median, interval and probabilities odhad; investment in the base case výpočet. At 6 % and 10 % the verdict is the same (T2, T3): the deadline probability does not depend on the rate. Keep buying costs 897 mil. EUR in the median of the futures and 911,6 in the base case.

B.2 T4 · Finance, base case

Site	Pre-tax 6 %	8 %	10 %	Post-tax 8 %	Median post-tax 8 %	IRR	Payback	Start (base)
Vrbice	189,2	134,5	92,7	96,6	87,5	18,2 %	2035	2030,92
Lipová	219,3	159,9	113,8	116,3	99,9	19,2 %	2034	2029,92
Kamenec	225,6	165,9	119,6	121,3	107,3	19,9 %	2034	2029,92
Wendelstedt	149,8	99,8	61,2	84,3	72,1	14,8 %	2035	2029,92
Lindenau	156,0	105,7	67,0	92,4	80,0	15,5 %	2035	2030,17
Eschfeld	139,4	91,2	54,2	69,3	45,9	14,4 %	2036	2030,25
Odrzyca	230,6	170,7	124,2	118,6	107,2	20,5 %	2034	2029,92
Bory	200,5	145,8	103,9	99,7	96,0	20,1 %	2035	2031,17

Base case výpočet; the post-tax median odhad. Tax rates provisional. Incentives are not in the base case; the futures grant them with probability 0,5.

B.3 T6 · Operating cost per tonne in 2032, base case, EUR/t

Site	CO_2	Freight	Energy	People	Waste	Raw materials	Water	Maintenance, park	Total
Vrbice	56	38	179	121	17	760	26	113	1 311
Lipová	56	20	179	116	20	760	25	124	1 300
Kamenec	56	24	179	124	16	760	26	111	1 295
Wendelstedt	56	58	182	203	13	760	25	157	1 455
Lindenau	56	53	190	195	15	760	32	132	1 434
Eschfeld	56	55	190	232	14	760	34	122	1 463
Odrzyca	56	37	192	102	18	760	21	105	1 292
Bory	56	33	192	109	17	760	20	103	1 290

výpočet. A bought tonne would cost about 2 340 EUR in 2032 in the base case, premium and freight included.

B.4 T7 · Relations between sites across the futures

	Vrb	Lip	Kam	Wen	Lin	Esch	Odr	Bory
Vrbice	–	0,50	0,52	−0,27	−0,27	−0,26	−0,15	−0,07
Lipová	0,50	–	0,55	−0,22	−0,23	−0,27	−0,14	−0,15
Kamenec	0,52	0,55	–	−0,23	−0,23	−0,26	−0,14	−0,15

	Vrb	Lip	Kam	Wen	Lin	Esch	Odr	Bory
Wendelstedt	−0,27	−0,22	−0,23	–	0,56	0,70	−0,12	−0,17
Lindenau	−0,27	−0,23	−0,23	0,56	–	0,71	−0,12	−0,15
Eschfeld	−0,26	−0,27	−0,26	0,70	0,71	–	−0,15	−0,15
Odrzyca	−0,15	−0,14	−0,14	−0,12	−0,12	−0,15	–	0,48
Bory	−0,07	−0,15	−0,15	−0,17	−0,15	−0,15	0,48	–

`odhad`; `aia-sociomap-5`, the judgement's correlations, centred per future. Two of 28 pairs reliable. Split-half 0,91 against 0,59. Stability: 0,95 / 0,84 / 0,75 between the three settings, the last under 0,8: failed.

B.5 T8 · Shared failure of the two leading sites

When this site is worse than buying …	Vrb	Lip	Kam	Wen	Lin	Esch	Odr	Bory
Odrzyca	90 %	84 %	79 %	98 %	97 %	99 %	–	91 %
Kamenec	99 %	93 %	–	98 %	98 %	100 %	75 %	83 %

`odhad`; shares of the futures, uncentred. No site is a backup of another.

B.6 The decision boundary and the input table in brief

Compensation chance	Rule chooses	P(on time) Lipová / Kamenec / Odrzyca
0 %	Odrzyca	0,942 / 0,942 / 0,942
10 %	Odrzyca	0,852 / 0,851 / 0,844
14 %	Odrzyca	0,817 / 0,813 / 0,808
15 %	Kamenec	0,807 / 0,804 / 0,797
16 %	nothing	0,797 / 0,795 / 0,787
20 % (the model)	nothing	0,760 / 0,759 / 0,748

`odhad` (R2-P06 § 5.3). T9 holds 54 inputs: 16 public inputs await verification (*prozatímní*), 22 inputs the files do not hold are judgements (*úsudek*), the rest are read from client data (*ověřeno*); a public input can also be a judgement, so the groups overlap. Overrun and delay: median 17 %, 90th percentile 30,2 %, delay median 3 months, from 36 comparable plants, drawn together.

APPENDIX C

Board index

The thirty boards of the Design canvas *AIA Obchodní model*, version 51, in the order of its rows, with the session that changes each in this round (the brief's session table). "In revision" means another session is changing the board while this document is written; the coordinator folds the result in. Boards marked parked stay on the canvas as planned, not deleted.

Code	Board	File	Session	State in this round
Z0	*Zadání: průchod*	`Z0Flow.dc.html`	A	in revision: the flow and the board list around the spine
Z1	*Zadání: začátek*	`Z1Start.dc.html`	—	unchanged
Z2	*Zadání: triáž*	`Z2Triage.dc.html`	—	unchanged
Z3	*Zadání: rozhodnutí a pravidlo*	`Z3Decision.dc.html`	B	in revision: the type-2 rule with four conditions and the interpretation record
Z4	*Zadání: analýzy do zprávy*	`Z4Analyses.dc.html`	—	unchanged
Z5	*Zadání: obchodní model*	`Z5Model.dc.html`	B	in revision
Z7	*Zadání: připravenost a předání*	`Z7Ready.dc.html`	—	unchanged
M1	*Model: Data*	`KM1Data.dc.html`	B	in revision: contracts, the findings table P1 with dispositions
MV1	*Data: datová smlouva*	`MV1Contract.dc.html`	B	in revision: units, currencies, price years
MV2	*Rešerše: typované vstupy*	`MV2PublicData.dc.html`	B	in revision: public inputs with state and verification at source
MV2b	*Rešerše: zjištění podle analýz*	`MV2bFindings.dc.html`	—	unchanged
M2	*Model: Předpoklady*	`KM2Assumptions.dc.html`	B	in revision: the T9 table, estimates from comparables, scope check, correlations with the stability test declared
M3	*Model: Model*	`KM3Model.dc.html`	C	in revision: timeline, cost of supply, base case, futures, stopping rule
M4	*Model: Výsledky*	`KM4Results.dc.html`	C	in revision: the rule table at three rates, the boundary, delays, tornado, break-even, value of information
M4b	*Výsledky: Business case a finance*	`KM7Finance.dc.html`	C	in revision
M4c	*Výsledky: Účetnictví*	`KM8Accounting.dc.html`	C	parked for type 2
M4d	*Výsledky: Operační výzkum*	`KM9Operations.dc.html`	C	in revision: cost per tonne by category
M4f	*Výsledky: Logistika*	`KM11Logistics.dc.html`	—	unchanged
M4e	*Výsledky: Velikost trhu*	`KM10Market.dc.html`	C	parked for type 2
M4g	*Výsledky: Harmonogram*	new	C	new: start-date distributions against the deadline
MV3	*Výsledky: Sociomapa systémů*	`MV3SystemMap.dc.html`	D	in revision: the futures map as the exemplar draws it, T7, T8, the stability test
MV9	*Mapa scénářů vah*	new	D	new: the weighting-scenario map on the real engine
SI1–SI4	*Sociomapa z indikátorů*: method, matrix, options map, suppliers map	`SI1Method` … `SI4Suppliers.dc.html`	D	in revision: reduced to what MV9 needs
MV4	*Osnova zprávy*	`MV4XRay.dc.html`	E	in revision: the part structure R2-P01…P18 and the grade and source conventions
MV8	*Zpráva: kontrola platnosti modelu*	`MV8Validity.dc.html`	E	in revision: the exemplar's admitted limits
M6	*Model: Zpráva*	`KM6Report.dc.html`	E	in revision: the report AIA renders

Code	Board	File	Session	State in this round
MV0	*Definované MVP: přehled*	`MV0Scope.dc.html`	A	in revision: rewritten around the spine; acceptance = the answer key
MV5	*Metodika: validační sada a otázky*	`MV5Validation.dc.html`	E	in revision: the register entries the exemplar waits on
MV6	*Katalog analýz*	`MV6Catalogue.dc.html`	—	unchanged; the catalogue beyond the perspectives is parked

Sizes and positions: `canvas.json`. Reference renders of version 51: `reference/`. The canvas is published by the coordinator from the exported files; no session publishes it.